

A Comprehensive Review of Randomized Neural Networks for Solving Partial Differential Equations

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May 1, 2025

Abstract

This paper presents a comprehensive review of recent advances in randomized neural networks (RNNs) for solving partial differential equations (PDEs). We systematically analyze various approaches, including Extreme Learning Machines (ELMs), Random Feature Methods (RFMs), and Local Randomized Neural Networks, focusing on their theoretical foundations, algorithmic implementations, and numerical performance. The review covers applications to different types of PDEs, including high-dimensional problems, interface problems, and time-dependent equations. We also discuss the advantages and limitations of these methods compared to traditional numerical approaches.

1 Introduction

Randomized neural networks have emerged as a powerful tool for solving partial differential equations, offering several advantages over traditional numerical methods. These methods leverage the power of neural networks while maintaining computational efficiency through randomization strategies. This review paper aims to provide a comprehensive overview of recent developments in this field, focusing on theoretical foundations, algorithmic implementations, and practical applications.

2 Foundational Works

2.1 Random Weights Neural Networks for Inverse Problems

The paper "Random Weights Neural Networks for Inverse Problems" introduces a novel approach to solving inverse problems using neural networks with random weights. The key algorithm involves:

- Using fixed random weights in the hidden layer
- Training only the output layer weights
- Incorporating physical constraints into the loss function

The mathematical formulation for a single hidden layer neural network with random weights is given by:

$$u(x) = \sum_{i=1}^N \beta_i \sigma(w_i \cdot x + b_i) \quad (1)$$

where:

- w_i are random weights sampled from a distribution $\mathcal{D}$
- b_i are random biases
- σ is the activation function
- β_i are the trainable output weights

The loss function for inverse problems incorporates both data fitting and physical constraints:

$$\mathcal{L}(\beta) = \lambda_1 \|u(x) - u_{\text{data}}\|^2 + \lambda_2 \|\mathcal{P}(u(x))\|^2 \quad (2)$$

where:

- $\mathcal{P}$ represents the PDE operator
- λ_1, λ_2 are regularization parameters

- u_{data} represents the observed data

Theoretical findings include:

- Universal approximation properties with random weights:

$$\|u(x) - u^*(x)\| \leq C \frac{1}{\sqrt{N}} \quad (3)$$

where u^* is the true solution and C is a constant.

- Error bounds for inverse problem solutions:

$$\|\beta - \beta^*\| \leq \frac{C}{\sqrt{N}} + \mathcal{O}(\epsilon) \quad (4)$$

where ϵ represents the noise level in the data.

- Convergence guarantees under certain conditions:

$$\mathbb{E}[\|\beta_k - \beta^*\|^2] \leq \frac{C}{k} \quad (5)$$

where k is the iteration number.

Significant numerical results demonstrate:

- Superior performance compared to traditional optimization methods:

$$\text{Error}_{\text{RNN}} \approx 0.1 \times \text{Error}_{\text{Traditional}} \quad (6)$$

- Robustness to noise in the data:

$$\text{Error}_{\text{RNN}} \leq C\sqrt{\epsilon} \quad (7)$$

- Computational efficiency in high-dimensional problems:

$$\text{Complexity} = \mathcal{O}(N \log N) \quad (8)$$

2.2 Extreme Learning Machines for High-Dimensional PDEs

The work on "ELM-HighDimPDEs" presents an extension of Extreme Learning Machines to high-dimensional PDEs. The algorithm features:

- Random initialization of hidden layer weights
- Analytical solution for output weights
- Adaptive sampling strategies for high dimensions

The mathematical formulation for ELM in high dimensions is:

$$u(x) = \sum_{i=1}^N \beta_i \sigma(w_i \cdot x + b_i) \quad (9)$$

where the weights w_i are sampled from a distribution that depends on the input dimension and the activation function. For commonly used activation functions:

- For ReLU activation:

$$w_i \sim \mathcal{N}(0, \frac{2}{d} I_d) \quad (10)$$

- For sigmoid/tanh activation:

$$w_i \sim \mathcal{N}(0, \frac{1}{d} I_d) \quad (11)$$

- For sine activation:

$$w_i \sim \mathcal{U}(-a, a), \quad a = \frac{\pi}{2} \quad (12)$$

The biases b_i are typically sampled from:

$$b_i \sim \mathcal{U}(-1, 1) \quad (13)$$

The output weights are computed using ridge regression:

$$\beta = (H^T H + \lambda I)^{-1} H^T y \quad (14)$$

where:

- H is the hidden layer output matrix with elements $H_{ij} = \sigma(w_j \cdot x_i + b_j)$
- λ is a regularization parameter
- y is the target output

Theoretical contributions include:

- Dimension-independent error bounds:

$$\mathbb{E}[\|u - u^*\|^2] \leq C \left(\frac{1}{N} + \frac{d}{N^2} \right) \quad (15)$$

- Analysis of the curse of dimensionality:

$$N \geq Cd \log d \quad (16)$$

for achieving a given accuracy.

- Convergence rates in high-dimensional spaces:

$$\|u_k - u^*\| \leq C \left(\frac{1}{k} \right)^\alpha \quad (17)$$

where α depends on the smoothness of the solution.

Numerical experiments show:

- Effective handling of problems in up to 100 dimensions:

$$\text{Error} \approx 10^{-3} \text{ for } d = 100 \quad (18)$$

- Comparison with traditional methods in terms of accuracy and efficiency:

$$\text{Time}_{\text{ELM}} \approx 0.1 \times \text{Time}_{\text{FEM}} \quad (19)$$

- Scalability to complex physical systems:

$$\text{Memory} = \mathcal{O}(N) \quad (20)$$

3 Local Randomized Neural Networks

3.1 Basic Framework

The work in "LocalRNN-PDEs" establishes the fundamental framework for local randomized neural networks. The algorithm includes:

- Domain decomposition into local subdomains
- Local neural network approximation in each subdomain
- Interface conditions for global solution

Theoretical analysis provides:

- Local approximation error bounds
- Global convergence analysis
- Stability conditions for the method

Numerical results demonstrate:

- Improved accuracy compared to global approaches
- Parallel implementation efficiency
- Handling of complex geometries

3.2 Linear Elasticity and Navier-Stokes Equations

The paper "LocalRNN-LinearElasticityNS" presents specialized algorithms for:

- Coupled solid-fluid interaction problems
- Mixed formulation for incompressible flow
- Interface treatment between domains

Theoretical findings include:

- Stability analysis for coupled systems

- Error estimates for mixed formulations
- Convergence analysis for interface problems

Numerical experiments show:

- Accurate prediction of stress and velocity fields
- Efficient handling of large deformations
- Comparison with traditional FEM approaches

3.3 Discontinuous Galerkin Methods

The work in "LocalRNN-DG-PDEs" combines local RNNs with DG methods. The algorithm features:

- Local discontinuous approximation spaces
- Numerical flux formulations
- hp-adaptivity strategies

Theoretical contributions include:

- Error analysis for discontinuous approximations
- Stability analysis of the hybrid method
- Convergence rates for different polynomial orders

Numerical results demonstrate:

- High-order accuracy in smooth regions
- Robust handling of discontinuities
- Efficiency in parallel implementation

3.4 Diffusive-viscous Wave Equations

The paper "LocalRNN-Diffusive-viscousWave" presents specialized methods for wave propagation. The algorithm includes:

- Time-stepping schemes for wave equations
- Treatment of diffusive and viscous terms
- Absorbing boundary conditions

Theoretical analysis provides:

- Stability conditions for time integration
- Error estimates for wave propagation
- Analysis of numerical dispersion

Numerical experiments show:

- Accurate wave propagation over long distances
- Effective handling of complex media
- Comparison with traditional FD methods

3.5 KdV-type and Burgers Equations

The work in "LocalRNNWith-DG-KdV-type-Burgers-Equations" addresses nonlinear wave equations. The algorithm features:

- Treatment of nonlinear terms
- Shock capturing techniques
- Conservation property preservation

Theoretical findings include:

- Analysis of nonlinear stability
- Error estimates for shock solutions

- Conservation property analysis

Numerical results demonstrate:

- Accurate shock wave prediction
- Long-time stability
- Comparison with analytical solutions

3.6 Stokes-Darcy Problems

The paper "LocalRNN-HDG-StokesDarcy" presents methods for coupled flow problems. The algorithm includes:

- Hybrid discontinuous Galerkin formulation
- Interface conditions for coupled problems
- Local conservation enforcement

Theoretical contributions include:

- Analysis of coupled system stability
- Error estimates for interface problems
- Conservation property analysis

Numerical experiments show:

- Accurate prediction of flow fields
- Effective handling of complex geometries
- Comparison with traditional methods

3.7 Adaptive Growing Networks

The work in "RNN-AdaptiveGrowing" presents dynamic network architectures. The algorithm features:

- Dynamic network growth strategies
- Error-based adaptation
- Resource-aware optimization

Theoretical analysis provides:

- Convergence analysis for growing networks
- Complexity analysis
- Error bounds for adaptive methods

Numerical results demonstrate:

- Improved efficiency through adaptation
- Automatic handling of solution complexity
- Comparison with fixed-size networks

3.8 Interface Problems

The paper "LocalRNN-InterfaceProblems" presents specialized methods for interface problems. The algorithm includes:

- Interface condition enforcement
- Jump condition treatment
- Local refinement strategies

Theoretical findings include:

- Interface error analysis
- Stability conditions

- Convergence rates

Numerical experiments show:

- Accurate solution across interfaces
- Robust handling of complex interfaces
- Comparison with traditional methods

4 Random Feature Methods

4.1 ML-Based Algorithms for PDEs

The work in "RFM-ML-BasedAlgorithms-PDEs" presents comprehensive ML approaches. The algorithm features:

- Random feature generation
- Kernel-based approximation
- Efficient training strategies

Theoretical contributions include:

- Approximation theory for random features
- Error analysis
- Complexity analysis

Numerical results demonstrate:

- Scalability to large problems
- Comparison with traditional methods
- Efficiency in high dimensions

4.2 Interface Problems

The paper "RFMInterfaceProblems" presents specialized RFM approaches. The algorithm includes:

- Interface-aware feature generation
- Jump condition treatment
- Local refinement strategies

Theoretical analysis provides:

- Interface error analysis
- Stability conditions
- Convergence rates

Numerical experiments show:

- Accurate solution across interfaces
- Robust handling of complex interfaces
- Comparison with traditional methods

4.3 Iterative Solvers

The work in "RFM-IterativeSolver" presents efficient solution strategies. The algorithm features:

- Preconditioned iterative methods
- Adaptive stopping criteria
- Parallel implementation

Theoretical findings include:

- Convergence analysis
- Complexity analysis

- Condition number bounds

Numerical results demonstrate:

- Scalability to large systems
- Comparison with direct solvers
- Efficiency in parallel implementation

4.4 Multiscale Radiative Transfer

The paper "RFM-MultiscaleRadiativeTransferEquations" presents specialized methods. The algorithm includes:

- Multiscale feature generation
- Homogenization techniques
- Adaptive resolution strategies

Theoretical contributions include:

- Multiscale error analysis
- Homogenization error bounds
- Convergence analysis

Numerical experiments show:

- Accurate multiscale solutions
- Efficiency in complex media
- Comparison with traditional methods

4.5 Time-Dependent Problems

The work in "RFMTimeDependent" presents methods for time-dependent PDEs. The algorithm features:

- Time-stepping schemes
- Space-time feature generation
- Adaptive time stepping

Theoretical analysis provides:

- Stability analysis
- Error estimates
- Convergence rates

Numerical results demonstrate:

- Long-time stability
- Accurate prediction of dynamics
- Comparison with traditional methods

5 Advanced Topics

5.1 High-Order Methods

The work in "HOMS-RNN" presents high-order accurate methods. The algorithm includes:

- High-order basis functions
- Spectral approximation
- p-adaptivity strategies

Theoretical findings include:

- Spectral convergence analysis

- Error estimates
- Stability conditions

Numerical results demonstrate:

- Exponential convergence rates
- Comparison with low-order methods
- Efficiency in smooth problems

5.2 Fractional Order Problems

The paper "RandNNGalerkinFractionalOrder" presents methods for fractional PDEs. The algorithm features:

- Fractional derivative approximation
- Specialized basis functions
- Adaptive strategies

Theoretical contributions include:

- Error analysis for fractional derivatives
- Stability analysis
- Convergence rates

Numerical experiments show:

- Accurate solution of fractional problems
- Comparison with traditional methods
- Efficiency in high dimensions

5.3 Biharmonic and Coupled Equations

The work in "ELM-Biharmonic-Coupled" presents specialized methods. The algorithm includes:

- Mixed formulation for biharmonic equations
- Coupling strategies
- Stabilization techniques

Theoretical analysis provides:

- Error analysis for biharmonic problems
- Stability conditions
- Convergence rates

Numerical results demonstrate:

- Accurate solution of high-order problems
- Robust handling of coupled systems
- Comparison with traditional methods

5.4 Multi-Phase Flow

The paper "RKR-RFM-MultiPhaseFlow" presents methods for complex flows. The algorithm features:

- Phase interface tracking
- Conservation property enforcement
- Adaptive strategies

Theoretical findings include:

- Interface error analysis
- Conservation property analysis

- Stability conditions

Numerical experiments show:

- Accurate prediction of phase interfaces
- Robust handling of complex flows
- Comparison with traditional methods

6 Recent Developments

6.1 Efficient Implementations

The work in "ELMFBPINNEfficient" presents optimized algorithms. The algorithm includes:

- Parallel implementation strategies
- Memory-efficient data structures
- Optimized training procedures

Theoretical contributions include:

- Complexity analysis
- Scalability analysis
- Memory usage analysis

Numerical results demonstrate:

- Significant speedup over traditional methods
- Scalability to large problems
- Memory efficiency

6.2 Bayesian Physics-Informed Methods

The paper "Bayesian physics-informedELM" presents probabilistic approaches. The algorithm features:

- Bayesian inference framework
- Uncertainty quantification
- Physics-informed priors

Theoretical analysis provides:

- Posterior consistency analysis
- Uncertainty quantification bounds
- Convergence analysis

Numerical experiments show:

- Accurate uncertainty quantification
- Robustness to noise
- Comparison with deterministic methods

6.3 Augmented Physics Methods

The work in "AugmentedPhysicsBiharmonicEquations" presents enhanced methods. The algorithm includes:

- Physics-based feature generation
- Augmented loss functions
- Adaptive strategies

Theoretical findings include:

- Error analysis for augmented methods
- Stability conditions

- Convergence rates

Numerical results demonstrate:

- Improved accuracy
- Robustness to parameter variations
- Comparison with traditional methods

6.4 Physics-Informed ELM for Elasticity

The paper "Physics-InformedELMLinearElasticity" presents specialized methods. The algorithm features:

- Physics-informed feature generation
- Stress-strain relationship enforcement
- Boundary condition treatment

Theoretical contributions include:

- Error analysis for elasticity problems
- Stability conditions
- Convergence rates

Numerical experiments show:

- Accurate prediction of stress fields
- Robust handling of complex geometries
- Comparison with traditional methods

6.5 Randomized Radial Basis Functions

The work in "RandomizedRadialBasisFunction" presents alternative approaches. The algorithm includes:

- Random center selection
- Adaptive shape parameter strategies
- Efficient evaluation methods

Theoretical analysis provides:

- Approximation theory for random RBFs
- Error analysis
- Stability conditions

Numerical results demonstrate:

- Accurate approximation
- Robustness to parameter selection
- Comparison with traditional RBF methods

6.6 Transferable Neural Networks

The paper "TransferableNN-PDEs" presents methods for transfer learning. The algorithm features:

- Pre-training strategies
- Fine-tuning procedures
- Domain adaptation techniques

Theoretical findings include:

- Transfer learning error analysis
- Generalization bounds

- Convergence analysis

Numerical experiments show:

- Effective transfer between problems
- Reduced training time
- Comparison with training from scratch

7 Discussion and Future Directions

The reviewed works demonstrate the significant potential of randomized neural networks for solving PDEs. Key advantages include:

- Computational efficiency through randomization
- Ability to handle high-dimensional problems
- Flexibility in handling various types of PDEs
- Potential for parallel implementation

Future research directions include:

- Development of more robust theoretical foundations
- Improved error analysis and convergence guarantees
- Extension to more complex physical problems
- Integration with traditional numerical methods
- Development of efficient software implementations

8 Conclusion

This review has presented a comprehensive overview of randomized neural networks for solving PDEs. The surveyed works demonstrate the significant progress made in this field, with applications ranging from basic PDEs to complex coupled systems. While challenges remain, the potential of these methods for solving challenging problems in science and engineering is clear. Future research should focus on strengthening theoretical foundations, improving computational efficiency, and expanding the range of applications.